

CAP404:INFORMATION SECURITY AND PRIVACY

L:4 T:0 P:0 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: define the fundamental security concepts and recognize importance of information security in the computer-driven world

CO2 :: explain the basics of cryptography and the working of cryptographic algorithms

CO3 :: demonstrate the underlying vulnerabilities of system and network Security

CO4 :: analyze the key privacy concepts in IT, principles, and requirements

Unit I

Introduction to information systems : basics of information systems, importance of information systems, the need for security, security approaches, principal of security, security services

Unit II

Cryptography-I : introduction to cryptography, role of cryptography in information systems, plain text and cypher text, key size and key range, encryption and decryption, symmetric and asymmetric key cryptography

Unit III

Cryptography-II : substitution techniques, caesar cipher, mono-alphabetic cipher, homophonic substitution cipher, polygram substitution cipher, transposition techniques, rail-fence technique, columnar transpositional technical, vernam cipher, book cipher

Unit IV

System security : intruders, viruses and related threats, the nature of viruses, types of viruses, types of security attacks, passive attack, active attack, application-level attack

Unit V

Network security : digital signature, firewalls, introduction, types of firewalls, design principles, characteristics, configuration, threats in networks, network attacks

Unit VI

Privacy in IT : introduction to privacy and information technology, privacy risks and harms, privacy law, policies and practices, surveillance, networks and privacy by design, privacy and government data, cyber risk, governing privacy in the university

Text Books:

1. CRYPTOGRAPHY AND NETWORK SECURITY by ATUL KAHATE, M.G.Hills

References:

1. INFORMATION SYSTEMS SECURITY - SECURITY MANAGEMENT, METRICS, FRAMEWORKS AND BEST PRACTICES SECOND EDITION by NINA GODBOLE, BHUVAN UNHELKAR, WILEY